



Towards railways safety: A systematic review on predictive diagnostics for axle bearings

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ABSTRACT

Axle bearings are critical for safe railway operation. Exposed to high mechanical loads and harsh environmental conditions, their failure can cause serious accidents. This systematic review analyses 75 studies on predictive diagnostics of axle bearings, published between 2014 and 2024. It focuses on methods such as vibration analysis, acoustic emissions, temperature monitoring, and AI-driven fault detection. The strengths and weaknesses of each approach are discussed, with attention to challenges like noise interference, real-world deployment, and the scalability of AI models. The review highlights the need for standardized validation procedures and better integration with predictive maintenance frameworks. Its goal is to support the development of reliable, real-time monitoring systems that enhance safety, reduce costs, and extend bearing life.

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1. Introduction

The necessity for predictive analysis of axle bearings in railway systems is underscored by the critical role these components play in ensuring operational safety and efficiency. Axle bearings are subjected to extreme conditions, including high speeds and heavy loads, which can lead to premature failure if not monitored effectively [1,2]. Predictive maintenance strategies, leveraging advanced data analytics

and machine learning techniques, have emerged as essential tools for enhancing the reliability of these components.

Unlike general industrial bearings, axle bearings in railway vehicles operate under far more demanding conditions, including heavy dynamic loads, high-speed rotation, strong environmental variability, and restricted accessibility for manual inspection [3]. These bearings are critical for safe railway operations, as their failure can result in severe consequences such as derailments. Their integration into the

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Table 1
Overview of related surveys.

Reference	Year	Title
Smith et al. [14]	2015	Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study
Neupane et al. [15]	2020	Bearing fault detection and diagnosis using case western reserve university dataset with deep learning approaches: A review
Zhang et al. [16]	2021	Machine Learning Based Bearing Fault Diagnosis Using the Case Western Reserve University Data: A Review
Yan et al. [17]	2022	A Survey on Fault Diagnosis Approaches for Rolling Bearings of Railway Vehicles
Hu et al. [18]	2023	Vibration-based bearing fault diagnosis of high-speed trains: A literature review

bogie structure and their exposure to vibrations and shocks from the track significantly differentiate them from bearings used in static or less mechanically complex environments. Therefore, axle bearings require specialized diagnostic approaches that account for these unique operational challenges, which justifies the dedicated focus of this review and highlights its practical relevance for railway safety [4].

One of the primary methods for diagnosing axle bearing conditions is vibration analysis. This technique has gained prominence due to its ability to detect early-stage faults that may not significantly affect temperature or produce noticeable acoustic signals. For instance, vibration data can reveal subtle changes in the bearing's operational state, allowing for timely interventions before a fault escalates into a more serious issue [5,6]. Studies have shown that vibration-based diagnostics can outperform other methods, such as temperature monitoring, particularly in the early stages of bearing degradation [7,8].

Moreover, advancements in diagnostic techniques, such as the use of convolutional neural networks and adaptive algorithms, have enhanced the accuracy of fault detection in axle bearings. These modern approaches leverage machine learning to analyze complex vibration signals, improving the identification of weak fault features that traditional methods might overlook [9,10]. The integration of big data analytics and condition-based maintenance strategies further supports proactive maintenance practices, maximizing the remaining useful life of bearings and minimizing unexpected failures [1,11].

Historical incidents show the critical importance of axle bearing diagnostics where bearing failures led to derailments, emphasizing the need for robust monitoring systems. For example, the derailment incidents in the United Kingdom, such as the Summit tunnel incident in 1984, highlighted the catastrophic consequences of undetected bearing faults, prompting the rail industry to invest heavily in condition monitoring technologies [12]. Continuous monitoring and diagnostics not only enhance safety but also contribute to operational efficiency by reducing downtime and maintenance costs associated with unexpected failures [2,13].

Table 1 presents related surveys on bearing fault detection. Smith et al. Neupane et al. and Zhang et al. review studies that use the Case Western Reserve University dataset [14–16]. Yan et al. examines the research progress in rolling bearing fault diagnosis technology for railway vehicles [17]; however, this review does not consider in-field deployments. Hu et al. [18] provides literature review, however consider only high-speed trains and vibration diagnostics.

This study aims to consolidate the current body of research on axle bearing diagnostics, focusing specifically on algorithms and methods for fault detection and wear assessment, as well as the fundamental principles enabling the evaluation of axle bearing condition. Furthermore, it identifies key research challenges within this scientific domain, providing a foundation for future advancements. Therefore this study defines the following contributions:

1. This review synthesizes state-of-the-art methodologies for bearing fault detection, including vibration analysis, acoustic emissions, and temperature monitoring.
2. It systematically compares traditional signal processing techniques with AI-driven methods, evaluating their effectiveness, scalability, and real-world applicability.
3. The study identifies key research gaps, such as the predominance of laboratory-based validation, deployment challenges, noise interference, and the lack of standardized testing.

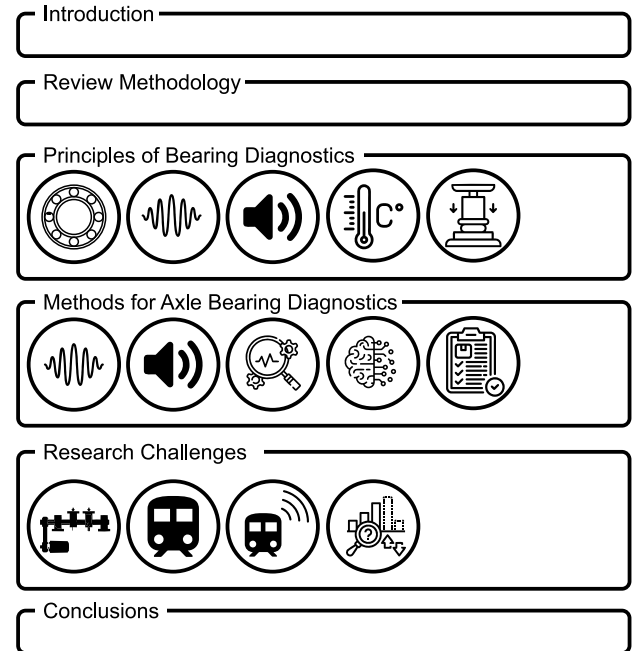


Fig. 1. Structural outline of the systematic review on predictive diagnostics for bearing.

4. A structured methodology is proposed, integrating laboratory experiments, field data analysis, and real-time predictive maintenance to bridge the gap between research and practical railway applications.

The rest of the article is organized as follows (see Fig. 1). Section 2 describes the review methodology, detailing the systematic selection process following the PRISMA framework to ensure a rigorous and unbiased literature review. Section 3 introduces the principles of bearing diagnostics, providing an overview of key fault detection methods, including vibration analysis, temperature monitoring, and acoustic emissions. Section 4 presents methods for bearing diagnostics, discussing traditional approaches, AI-based techniques, and hybrid methodologies that integrate multiple diagnostic principles. Section 5 highlights research challenges, including the lack of real-world validation, generalization difficulties of AI-based methods, regulatory compliance barriers, and the absence of comprehensive diagnostic frameworks integrating laboratory, field, and real-time deployment stages. Finally, Section 6 concludes the paper by summarizing key findings and outlining directions for future research in predictive maintenance for axle bearings in railway applications.

2. Review methodology

To ensure a comprehensive and systematic evaluation of the available literature on bearing diagnostics in railway systems, the review methodology adhered to the PRISMA framework.

The systematic review process which is shown in Fig. 2 began with a comprehensive search of the Scopus and Web of Science database. The search was defined using the query “bearing AND railway AND

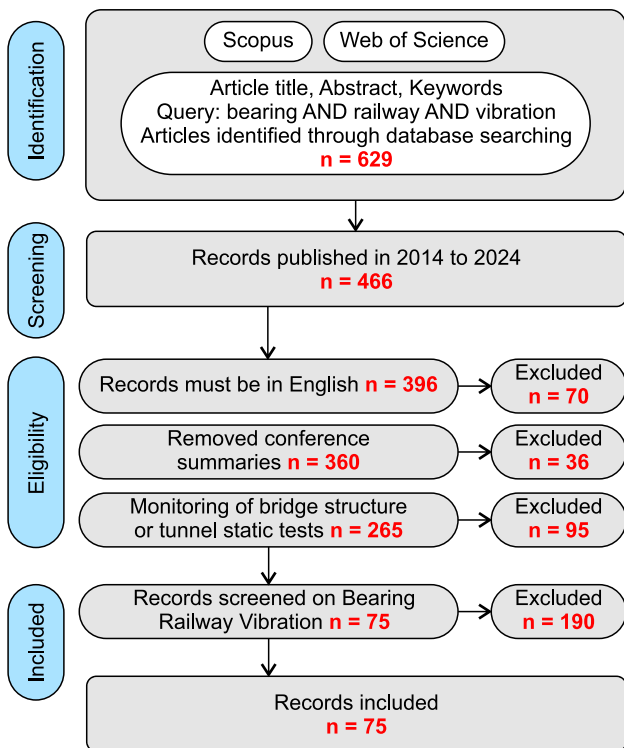


Fig. 2. PRISMA flow diagram of the systematic review process, showing the selection of 75 articles from an initial 629 Scopus records based on predefined inclusion and exclusion criteria.

vibration”, applied to article titles, abstracts, and keywords. The initial search yielded 629 records. During the screening phase, the records were limited to publications from 2014 to 2024, reducing the dataset to 466 articles. Non-English articles were subsequently excluded, leaving 396 records. Further filtering removed 36 conference summaries records and 95 articles referencing monitoring of bridge structure or tunnel static tests, resulting in 265 records. These remaining articles were then screened for relevance to bearing, railway, and vibration topics. References that were unavailable for access were also excluded, leaving a final dataset of 75 articles for detailed analysis.

3. Principles of bearing diagnostics

Bearing diagnostics rely on fundamental physical principles to detect early-stage faults and ensure railway safety. Key methods include vibration analysis, temperature monitoring, acoustic emissions, and mechanical stress evaluation. This section outlines these techniques, highlighting their advantages, limitations, and applications.

Axle bearings are essential mechanical components in railway vehicles, facilitating smooth motion and efficient load transfer between the wheels and the vehicle frame. Their primary role is to minimize friction and wear during rotational movement, ensuring reliable operation even under extreme mechanical and environmental conditions [2,6,7].

The design and functionality of axle bearings are influenced by various factors, including load conditions, environmental stresses, and the mechanical properties of the materials used. Axle bearings must withstand high loads and dynamic forces, which can lead to wear and potential failure if not properly monitored [17,19]. Studies have shown that the dynamic response of axle bearings can be affected by factors such as wheelset flexibility and wear patterns, which can exacerbate vibration and contact forces within the bearing assembly [19,20]. This highlights the importance of robust condition monitoring systems that can detect early signs of wear or damage, thereby preventing catastrophic failures [1,12].

Ball bearings (Fig. 3a) are generally less suitable for heavy axle systems due to their limited capacity to manage high radial and axial loads effectively. Their design, which typically features smaller rolling elements, restricts their load-bearing capabilities, making them more appropriate for lighter applications where the load and speed requirements are not as demanding [13,21]. In contrast, cylindrical roller bearings (Fig. 3b) use cylindrical rolling elements that facilitate a more extensive contact area, allowing for an even distribution of radial loads. This characteristic renders them particularly adept at handling high loads and speeds [7,22]. Tapered roller bearings (Fig. 3c) are extensively employed in railway applications due to their unique ability to accommodate combined radial and axial loads. This capability is crucial during operations that involve curves or dynamic loads resulting from track irregularities, where the bearings must effectively manage forces from multiple directions [8,12]. The tapered design of these bearings allows them to handle significant thrust loads while maintaining stability and performance, which is essential for the safety and reliability of railway vehicles [23].

Fig. 4 shows the primary types of failures in rolling element bearings: inner ring defects, outer ring defects, and defects in the rolling elements themselves, such as balls or tapered rollers. Each type of defect has distinct characteristics and implications for the operation of the bearing. Inner ring defects (Fig. 4a) typically manifest as surface damage or pitting on the raceway of the inner ring. These defects can arise from various factors, including improper lubrication, contamination, or excessive loads. The presence of an inner ring defect can lead to increased vibration levels and noise, which are critical indicators of bearing health [13,23]. The failure of the inner ring can also result in misalignment and uneven load distribution, exacerbating wear on other components and potentially leading to catastrophic failure if not detected early [17,23]. Outer ring defects (Fig. 4b) are characterized by similar surface damage or pitting on the raceway of the outer ring. These defects can significantly affect the overall performance of the bearing, as they can lead to increased radial play and misalignment, which in turn can cause premature failure of the rolling elements [24, 25]. Defects in the rolling elements, such as balls or tapered rollers, can occur due to fatigue, improper lubrication, or contamination. These defects can lead to uneven load distribution and increased stress on the bearing, resulting in accelerated wear and potential failure [7,26].

Fig. 5 presents examples of real-world bearing damage that are considered unacceptable for continued operation. These cases, observed during operational inspections or post-failure analyses, represent critical failure modes commonly encountered in rolling element bearings. Figure (a) shows severe corrosion and oxidation discoloration on the rolling elements, typically caused by lubricant degradation and moisture contamination. Such damage significantly compromises bearing functionality and is usually the result of inadequate sealing or prolonged exposure to humid environments. In Figure (b), surface fatigue, pitting, and oxidation are evident on the rollers, indicating excessive heat generation due to insufficient or degraded lubrication. These conditions promote the formation of microcracks, which propagate under repeated loading and lead to progressive surface deterioration. Figure (c) highlights localized mechanical damage on the raceway of the inner ring, likely caused by improper handling, installation errors, or contamination by hard particles. Although seemingly minor, such scratches act as stress concentrators and can initiate surface fatigue. Finally, Figure (d) illustrates fretting corrosion in the bore of the inner ring, caused by improper shaft fit—such as deviations in roundness or surface geometry. The resulting micromovements generate oxidized wear debris that degrades the contact interface and reduces mechanical stability.

The failures in axle bearings result in vibrations, which, based on the mechanical properties of the bearing system, manifest as dominant frequencies. These frequencies can be determined using the following equations:

$$f_i = \frac{n}{2} \cdot \frac{r}{60} \left(1 + \frac{d \cdot \cos \alpha}{D} \right) \quad (1)$$

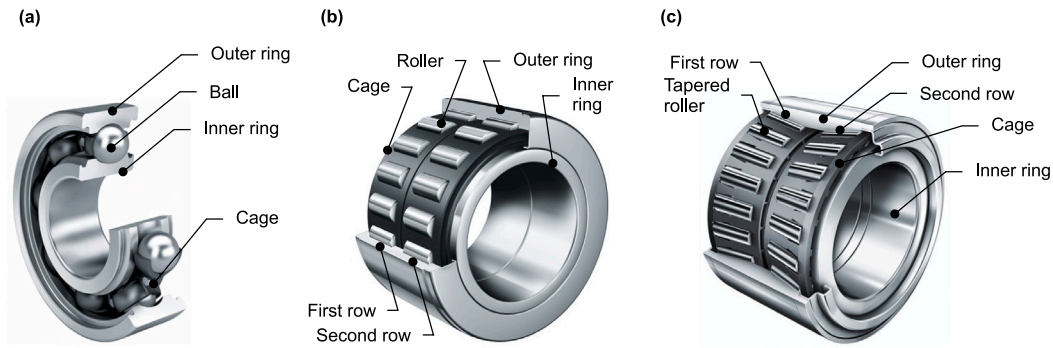


Fig. 3. Structural representations of bearings with labeled components: (a) Ball bearing, (b) Cylindrical roller bearing, (c) Tapered roller bearing.

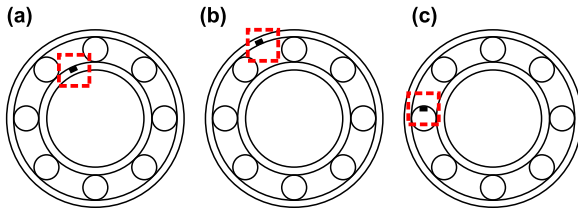


Fig. 4. Illustration of failure types in rolling element bearings: (a) inner ring defect, (b) outer ring defect, and (c) ball or tapered roller defect.

$$f_o = \frac{n}{2} \cdot \frac{r}{60} \left(1 - \frac{d \cdot \cos \alpha}{D} \right) \quad (2)$$

$$f_e = \frac{D}{d} \cdot \frac{r}{60} \cdot \left[1 - \left(\frac{d \cdot \cos \alpha}{D} \right)^2 \right] \quad (3)$$

where

- f_i ...fault frequency of inner ring (Hz)
- f_o ...fault frequency of outer ring (Hz)
- f_e ...fault frequency of element (Hz)
- n ...number of balls/rolling elements (-)
- r ...rotational speed (RPM)
- d ...element diameter (mm)
- D ...pitch diameter of axle bearing (mm)
- α ...contact angle (rad).

The equations provide a framework for identifying the dominant vibration frequencies corresponding to specific types of faults in axle bearings. The fault frequency of the inner ring (f_i) is derived based on the rotational speed (r), the number of rolling elements (n), and the bearing's geometric parameters, such as element diameter (d), pitch diameter (D), and contact angle (α). It characterizes the vibrational behavior induced by defects in the inner raceway [7,23]. Similarly, the outer ring fault frequency (f_o) accounts for these parameters, incorporating an additive term that reflects the outward force distribution on the outer raceway [2,5]. The rolling element fault frequency (f_e) introduces a squared term to capture the intricate dynamics of rolling elements under load and motion [24,27].

Recent studies have also explored the theoretical mapping between specific fault types and the resulting vibration responses in axle bearing systems under realistic operating conditions. For instance, it has been shown that raceway defects exhibit distinct dynamic signatures depending on the location (inner or outer ring) and the coupling between wheelset, suspension, and track irregularities [28]. Similarly, the influence of typical urban rail vehicle operations — such as curve negotiation and variable axle loads — on vibration and contact behavior of axle bearings has been systematically analyzed using vehicle-track coupled models [29]. These studies provide a theoretical basis for interpreting vibration features in field diagnostics and support the use of dynamic modeling to inform fault feature extraction strategies.

The diagnosis of bearing faults can be effectively achieved through various physical principles, including vibration measurement, temperature measurement, acoustic emissions, and mechanical stress analysis.

Vibration measurement is one of the most prevalent techniques for detecting early-stage bearing defects. It allows for the identification of characteristic frequencies associated with specific fault types [6,13,30], however, the challenge lies in distinguishing these signals from background noise, which can obscure fault signatures, particularly in the early stages of defect development [30].

Temperature measurement serves as another critical diagnostic tool, as an increase in temperature often indicates increased friction and wear within the bearing [21,31]. Continuous monitoring of temperature can provide early warnings of potential failures, allowing for timely maintenance interventions [8].

Furthermore, acoustic emission analysis has emerged as a valuable technique for condition monitoring, as it captures high-frequency stress waves generated by the rapid release of energy during crack formation and propagation within the bearing [8,32]. This method can detect faults that may not be apparent through vibration analysis alone, thus providing a complementary diagnostic approach.

Mechanical stress analysis, particularly through the application of finite element methods, can also contribute to fault detection by simulating the operational conditions of bearings and predicting failure modes based on stress distributions [17,19]. By integrating these various diagnostic principles, a more comprehensive understanding of bearing health can be achieved, facilitating the implementation of predictive maintenance strategies that enhance the reliability and safety of railway systems [1,11]. The combination of these methodologies not only improves fault detection capabilities but also aids in the assessment of the remaining useful life of bearings, ultimately contributing to more efficient railway operations.

The Table 2 provides a comprehensive summary of research initiatives dedicated to the application and testing of vibration-based diagnostic methods within controlled laboratory settings. It is noteworthy that all the laboratories mentioned rely exclusively on specialized test rigs, and there is a conspicuous absence of references indicating that these methods have been validated in real-world field conditions. This limitation raises questions about the practical applicability of the findings derived from such controlled environments.

In China, several leading laboratories are at the forefront of research focused on railway traction systems and bearing diagnostics. The State Key Laboratory of Traction Power in Chengdu is particularly noted for its emphasis on health monitoring and fault diagnostics of critical components in railway systems [5]. Additionally, the State Key Laboratory of Rail Traffic Control and Safety is advancing technologies aimed at fault diagnostics and condition monitoring for railway vehicles [37]. The Key Laboratory of Performance Guarantee on Urban Rail Transit Vehicles is also contributing significantly by enhancing the safety, reliability, and performance of urban rail systems through innovative diagnostic techniques [30]. Collectively, these institutions are making

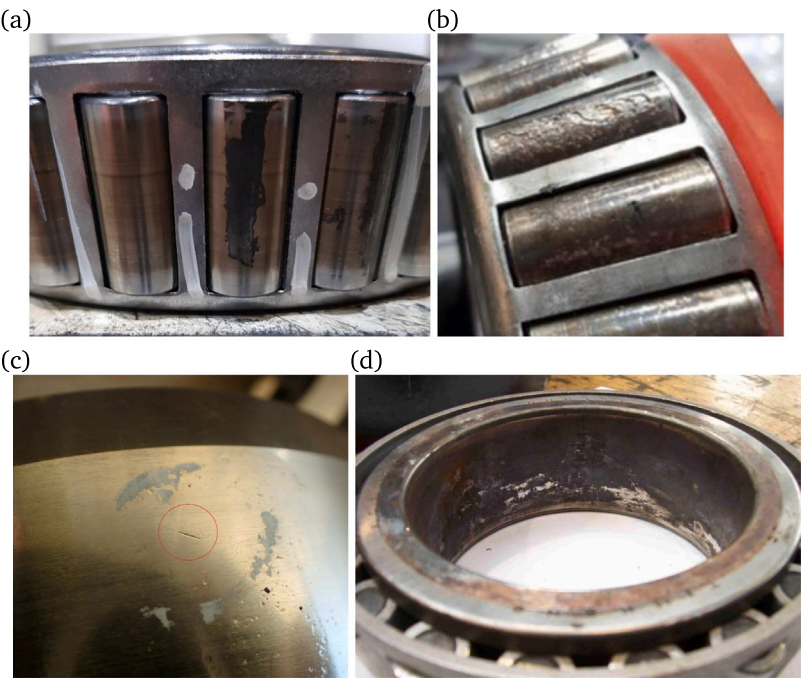


Fig. 5. Examples of real bearing damage identified during operational inspections:(a) severe corrosion and moisture-induced discoloration; (b) surface fatigue and microcracks from heat and lubrication failure; (c) localized mechanical damage to the inner raceway; and (d) fretting corrosion in the bore due to shaft-seat geometry issues.

Table 2
Overview of laboratories equipped with specialized test rigs for advanced vibration diagnostics of bearing systems.

Reference	Laboratory	Focus
[5,26,33–46]	State Key Laboratory of Traction Power, Southwest Jiaotong University, Chengdu, China	Advanced research and development in the field of railway traction systems, particularly the health monitoring and fault diagnosis of critical components.
[37,47–50]	State Key Laboratory of Rail Traffic Control and Safety	Advancing technologies and methodologies including fault diagnosis, condition monitoring, and the development of intelligent systems for railway vehicles.
[10,30]	Key Laboratory of Performance Guarantee on Urban Rail Transit Vehicles	Enhancing the safety, reliability, and performance of urban rail transit systems through advanced research in various aspects of railway vehicle technology.
[51,52]	State Key Lab of Rail Traffic Control and Safety	Develop innovative technologies and methodologies that enhance the operational performance of urban rail transit and high-speed rail systems.
[6,7,14–16,53–55]	Case Western Reserve University (CWRU) — Bearing Data Center, Seeded Fault Test Data	Extensive database of bearing fault data, which includes vibration signals from various types of bearings under different fault conditions.
[21,24,56]	University Transportation Center for Railway Safety	Enhancing the safety, reliability, and efficiency of railway systems in condition monitoring, fault diagnosis, and predictive maintenance of railway components.
[9,20,57–60]	Other laboratories in China	Advanced techniques for fault detection and dynamic analysis.
[8,25,61]	Laboratories in Europe	Condition monitoring and damage assessment, with a focus on defining damage indices and analyzing the influence of speed and defect sizes.
[62]	University of Queensland, Brisbane, Australia	Monitoring and modeling of false brinelling.

substantial contributions to the field, particularly in the areas of fault detection and dynamic analysis of bearing systems.

Internationally, Case Western Reserve University (CWRU) has established a comprehensive database of bearing fault data, which includes vibration signals from various bearing types tested under controlled conditions [6]. The University Transportation Center for Railway Safety is engaged in efforts to improve the safety and reliability of railway systems through predictive maintenance and vibration diagnostics [21]. European laboratories are similarly focused on condition monitoring and damage assessment, with particular attention to how speed and defect sizes influence diagnostic outcomes [8]. Furthermore, the University of Queensland in Brisbane, Australia, specializes in the monitoring and modeling of false brinelling in bearings, which is critical for understanding wear mechanisms in railway applications [62].

The Table 3 summarizes significant studies that provide in the field experiments, focusing on the contributions, deployment environments, and methods employed. These studies provide insights into vibration monitoring techniques, though they primarily emphasize parameter analysis or evaluation rather than a detailed classification of axle bearings.

Sun et al. [63] developed a framework to enhance detection properties by mitigating background noise in wheel wear-off using the Angle Domain Synchronous Averaging Technique. This work was conducted in both simulation and real train environments in China. Guan et al. [19] explored the relationship between axle flexibility, bearing dynamics, and vertical vibrations using a vehicle dynamic model, with research conducted in simulation on the Wuhan–Guangzhou railway. Milovancevic et al. [64] focused on experimentally validating the vibration spectrum of train ball bearings, using an adaptive neuro fuzzy inference system through measurements on actual trains.

Trilla et al. [65] proposed a low-resolution companding autoencoder to enable efficient vibration data compression. This method was tested in a field deployment on the London Northern Line underground system. Aravanis et al. [66] conducted spectral analysis of railway vehicle vertical vibrations using both non-parametric and parametric methods, with experiments carried out on the Athens metro. Montalvo et al. [21] analyzed vehicle vertical vibrations under normal operating conditions, employing a combination of non-parametric and parametric methods, with research performed on a railcar in Pueblo. Amini et al. [8] evaluated the effect of speed and defect size on high-frequency

Table 3

Overview of field research on vibration monitoring of axle bearings.

Reference	Contribution	Deployment	Method
Sun et al. [63]	Developed a framework that enhances detection properties by mitigating background noise in wheel wear-off	Simulation and real train in china	Angle Domain Synchronous Averaging Technique
Guan et al. [19]	describes the relationship between axle flexibility, bearing dynamics and vertical vibrations	Simulation/Wuhan-Guanghzou	Vehicle dynamic model
Milovancevic et al. [64]	Basics and experimentally validates the vibration spectrum of train ball bearings in motion.	Experimental measurement in train	Adaptive neuro fuzzy inference system
Trilla et al. [65]	Low-resolution companding autoencoder for efficient vibration data compression	London Northern Line underground	Neural network-based signal compression
Aravanis et al. [66]	Spectral analysis of railway vehicle vertical vibration	Athens metro	Non-parametric and parametric spectral analysis
Montalvo et al. [21]	Vehicle vertical vibrations under normal operating conditions using both non-parametric and parametric methods	Railcar in Pueblo	Vibration and temperature profiles analysis
Amini et al. [8]	Evaluation of the effect of speed and defect size on high frequency acoustic emission and vibration	Rail track in Long Marston	Parameters evaluation

Table 4

Physical principles used to diagnostics of axle bearing except vibration.

Reference	Contribution	Deployment	Method
Temperature			
Marquez et al. [11]	Condition based maintenance case studies	Case study on Kazakhstan's railways	Various machine learning methods
Liu et al. [2]	Real-time fault early warning method for high-speed electric multiple unit	Electrical multiple unit	Neural networks
Gu et al. [31]	High-speed train temperature anomaly diagnosis using multitask deep learning	Wuhan-Guangzhou line	Multitask deep learning
Yao et al. [67]	New approach uses infrared radiation measurements of bearings	Laboratory stand	machine learning (subdomain adaptation) retraining from different machine learning models
Acoustic emissions			
Ellis et al.[32]	The article validates the use of acoustic analysis to assess bearing condition	Laboratory stand	acoustic signal processing methods
Huang et al. [68]	mathematical model to describe the generated vibration frequencies of a damaged bearing	Laboratory stand and deployed train	mathematical methods of frequency analysis
Papaelias et al. [12]	Onboard condition monitoring system using high-frequency acoustic emission and vibration analysis	Portugal Lisboa-Cas-Cais	High-frequency acoustic emission analysis and vibration analysis
Mechanic stress			
Symonds et al. [13]	Describes the effect of vibration on the type of bearing damage	discarded bearings from deployed trains	non-destructive and destructive testing of damaged bearings (mechanical and chemical examination)
Wilson et al. [69]	Service load factor, a predictive tool for assessing the service	Laboratory experiments	X-ray diffraction analysis

acoustic emissions and vibrations, using parameter evaluation methods in a field study conducted on a rail track in Long Marston.

The Table 4 provides a structured overview of research studies that leverage alternative physical principles such as temperature, acoustic emissions, and mechanical stress for diagnosing axle bearings. These studies are systematically categorized based on their references, contributions, deployment contexts, and methodologies.

Temperature-based diagnostics have gained considerable attention and have been effectively deployed in various contexts, particularly for identifying bearings exhibiting severe damage. Márquez et al. [11] investigated condition-based maintenance through case studies on Kazakhstan's railways, using diverse machine learning methods to enhance predictive capabilities. Liu et al. [2] proposed a real-time fault early warning system for high-speed electric multiple units, employing neural networks to facilitate timely interventions. Additionally, Gu et al. [31] implemented multitask deep learning techniques for diagnosing temperature anomalies in high-speed trains. Yao et al. [67] introduced a novel approach that uses infrared radiation measurements of bearings, integrating machine learning with subdomain adaptation and model retraining. Despite these advancements, it is important to note that temperature-based methods are generally limited to detecting bearings

that have already sustained significant damage, thereby constraining their effectiveness in early fault detection.

Acoustic emission analysis serves as a complementary diagnostic approach, providing valuable insights into bearing conditions. Ellis et al. [32] validated the use of acoustic analysis for assessing bearing health through acoustic signal processing methods applied in laboratory settings. Huang et al. [68] developed a mathematical model to characterize the frequencies generated by damaged bearings, which was tested in both laboratory and field environments. Furthermore, Papaelias et al. [12] implemented an onboard condition monitoring system that combined high-frequency acoustic emission analysis with other diagnostics. This integration enhances the robustness of fault detection and allows for a more comprehensive understanding of bearing health.

Mechanical stress analysis has also been investigated as a diagnostic tool. Symonds et al. [13] examined the effects of vibration on bearing damage using discarded bearings from deployed trains, employing both non-destructive and destructive testing methods, including mechanical and chemical examination. Wilson et al. [69] introduced the service load factor as a predictive tool for assessing the service condition of bearings through laboratory experiments, leveraging X-ray diffraction analysis.

Table 5
Methods used for diagnostics of axle bearings faults.

Techniques	Key references	Limitations
Vibration-Based Methods		
• Time-Domain Analysis: RMS, Crest Factor, Kurtosis • Frequency-Domain Analysis: FFT, Envelope Spectrum • Time-Frequency Analysis: Wavelet Transform, EEMD, TVFEMD	[6,13,21,23–26,35,41,64,70–74]	Sensitive to noise and non-stationary signals. Performance varies with sensor placement. Computationally intensive for large datasets or complex signals.
Acoustic Methods		
• Trackside Acoustic Monitoring: Doppler Correction, Beamforming • Acoustic Emission (AE): Signal Envelope Analysis	[8,12,17,32,38,66,75,76]	Prone to environmental noise interference. Requires precise calibration and preprocessing to avoid false alarms.
Signal Processing		
• Decomposition Methods: EMD, EWT, SVD, TVFEMD • Filtering & Denoising: MOMEDA, Stochastic Resonance • Advanced Spectral Analysis: Spectral Kurtosis, Coherence, Resonance Demodulation	[5,6,9,20,30,31,33,36,42,45–47,51,58,77,78]	Relies on precise parameter tuning and preprocessing. Computational complexity is high for iterative methods. Limited generalization for overlapping faults or extreme noise.
AI/ML-Based Methods		
• Shallow Models: SVM, Decision Trees, Ensemble Methods • Deep Learning: CNN, LSTM, Hybrid Networks • Anomaly Detection: Autoencoders, One-Class SVM	[1,2,7,11,39,44,48–50,54–56,65,67,69]	Requires large, high-quality datasets for training. Computationally expensive. Performance may degrade under unseen operating conditions or noise-heavy data.
Morphological Methods		
• Morphological Filtering: MUW, GCMO, MMF • Feature Enhancement: Structural Elements, Morphological Operators	[38,44,45,52,53,59,79,80]	Requires expertise for selecting structural elements. High computational cost for multi-stage filtering and optimization. Limited validation for multi-fault and noisy environments.
Hybrid Methods		
• Integration Approaches: Combining Signal Processing, AI, and Simulation • Energy Harvesting & Wireless Monitoring: Electromechanical Systems, Sensor Networks • Wear and Dynamic Analysis: FEM, Contact Mechanics	[10,11,19,22,27,34,37,40,43,49,60–63,73,76,81],	High computational demand due to multi-method integration. Generalizability depends on data quality, sensor calibration, and model input parameters. Real-world noise and variability pose challenges for robust validation.

4. Methods for bearing diagnostics

Various methods have been developed to identify and classify bearing faults at an early stage, minimizing maintenance costs and reducing the risk of catastrophic failures. These methods can be broadly categorized into vibration-based techniques, acoustic emission analysis, signal processing approaches, and AI/ML-based methodologies.

Recent studies, such as [13,24,25], have demonstrated that integrating multiple diagnostic approaches enhances the robustness of fault detection in railway applications. The overview of diagnostic methods for axle bearings, including their key references and limitations, is summarized in Table 5.

Vibration analysis is one of the most commonly used techniques for bearing fault detection due to its high sensitivity to incipient faults. These methods rely on analyzing the dynamic response of axle bearings through various statistical, spectral, and hybrid methodologies. Time-domain features such as root mean square (RMS), kurtosis, crest factor, and peak-to-peak values [6,21,23] offer an initial assessment of fault severity. However, they often lack specificity in differentiating between fault types, making frequency-domain methods such as Fast Fourier Transform (FFT) and Power Spectral Density (PSD) necessary for extracting fault signatures at characteristic defect frequencies [24,26,41]. While these traditional techniques have been widely used, they struggle with non-stationary signals, leading to the adoption of advanced time–frequency methods such as Wavelet Transform (WT), Empirical Mode Decomposition (EMD), and Variational Mode Decomposition (VMD) [35,70–72]. These approaches allow localized time–frequency analysis and enhance the identification of transient fault characteristics, making them particularly useful for early-stage defect detection in high-speed railway applications [73,74].

Another effective approach to axle bearing monitoring is the analysis of acoustic emissions (AE). AE sensors detect high-frequency stress waves produced by defect propagation within the material, providing an early indication of surface degradation. Studies such as [8,17,32] have demonstrated the ability of AE techniques to capture microscopic crack formations before they become severe enough to generate

significant vibration anomalies. However, one major challenge of AE-based methods is their susceptibility to environmental noise, making precise sensor placement and advanced filtering essential to avoid false alarms [38,66]. Trackside monitoring systems incorporating Doppler correction and beamforming techniques have been implemented to improve AE signal clarity and provide real-time monitoring of passing railway vehicles [12,75,76]. In these cases, complementary signal processing methods such as spectral kurtosis (SK) and high-frequency resonance techniques (HFRT) [38,66] enhance AE signal quality, reducing the influence of external noise sources. Despite these advancements, practical deployment remains challenging due to complex operational environments and the need for robust feature extraction techniques.

To improve the accuracy and robustness of fault detection, modern research has focused on hybrid methods that integrate advanced signal processing with statistical optimization techniques. Approaches such as Minimum Entropy Deconvolution (MED) and Stochastic Resonance (SR) [20,58,78] have been widely used to enhance impulsive fault features while mitigating noise interference. Meanwhile, blind deconvolution techniques [6,46,58] combined with Generalized Statistical Indicators (GSI) have been employed to reconstruct meaningful fault signals from noisy environments. A more recent development involves the application of multiscale morphological filtering techniques [44, 45,52,59,79], which use structural elements to isolate transient impulsive signals, offering enhanced robustness against environmental interference. Furthermore, advanced envelope analysis using Squared Envelope Spectrum (SES) and Hilbert Transform (HT) [36,42] has been demonstrated to effectively highlight amplitude modulations related to specific bearing defects. These methods are crucial for the extraction of informative features that enable early fault detection, as supported by studies such as [77,82].

Artificial intelligence (AI) and machine learning (ML) have significantly advanced the automation of fault diagnosis systems. With the growing availability of railway condition monitoring data, AI-driven approaches have become increasingly viable for detecting and classifying axle bearing faults. Studies such as [1,2,7,56] highlight the

Table 6
Key findings related to the diagnostic methods for axle bearings.

Category	Key findings
Vibration-Based Methods	• Effective for detecting fault patterns and quantifying severity. • Time–frequency methods are noise-resilient and extract detailed features.
Acoustic Methods	• Real-time defect detection using AE signals. • Trackside systems effective for continuous monitoring.
Signal Processing	• Enhances weak fault features under noisy conditions. • Allows detailed fault signal decomposition.
AI/ML-Based Methods	• Improves classification accuracy and detects complex faults. • Deep learning captures high-dimensional features for early detection.
Morphological Methods	• Isolates impulsive fault features in noisy environments. • Effective for compound fault detection.
Hybrid Methods	• Combines strengths of multiple methods for robust fault diagnostics. • Enables real-time monitoring and predictive maintenance.

effectiveness of deep learning models, particularly Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, in extracting fault-related features from complex vibration signals. Supervised learning techniques [39,49,54] using Support Vector Machines (SVMs) and ensemble classifiers have also been widely adopted for fault classification. These models achieve high accuracy when trained on extensive datasets, but their performance often declines in real-world scenarios where labeled data is limited. To address this, researchers have explored semi-supervised and unsupervised learning techniques such as autoencoders, One-Class SVMs, and Gaussian Mixture Models (GMM) [44,48,50,65,67]. These approaches allow models to learn from both labeled and unlabeled data, improving generalization across diverse operating conditions. Feature engineering techniques such as Mel-Frequency Cepstral Coefficients (MFCCs) and Amplitude Modulation Spectrograms (AMS) [55,69] have also been used to enhance fault classification performance. Despite these advancements, AI-based methods still require extensive computational resources, rigorous hyperparameter tuning, and domain adaptation techniques for real-world deployment.

The key findings related to the diagnostic methods for axle bearings are summarized in Table 6. Vibration-based methods excel in detecting fault patterns and quantifying their severity, with time–frequency techniques demonstrating noise resilience and the ability to extract detailed features. Acoustic methods, including AE signals, enable real-time defect detection, making trackside systems particularly effective for continuous monitoring. Signal processing techniques enhance weak fault features under noisy conditions and provide detailed fault signal decomposition. AI/ML-based methods significantly improve classification accuracy, efficiently detecting complex faults, with deep learning capturing high-dimensional features for early detection. Morphological methods are adept at isolating impulsive fault features in noisy environments and effectively identifying compound faults. Finally, hybrid methods combine the strengths of multiple diagnostic approaches, enabling robust fault diagnostics and supporting real-time monitoring and predictive maintenance.

Despite the growing use of deep learning for bearing diagnostics, many proposed models exhibit poor generalization when applied to unseen field data. Most algorithms are trained on datasets with controlled or artificial fault scenarios, and their robustness to noise, sensor variability, or multi-fault conditions is rarely benchmarked. This limits the transferability of research prototypes to certified, safety-critical railway systems. There is a pressing need for domain adaptation techniques and cross-line validation protocols aligned with industrial standards.

5. Research challenges

This review presents several key observations that help define possible directions and future research challenges. While laboratory studies using vibration analysis and machine learning have yielded promising results, their direct applicability to real-world railway operations remains limited. A crucial research need is to validate these approaches

using in-field data that reflect the noise, variability, and operational complexity of deployed railway systems. Existing field studies primarily emphasize parameter evaluation rather than implementing comprehensive, validated diagnostic frameworks.

To address these issues, future research should prioritize the development of standardized protocols that bridge laboratory testing with real-world validation. This includes creating shared field datasets, establishing performance benchmarks under operational noise, and exploring fault generalization across multiple train configurations.

5.1. Emerging trends in methodology and predictive maintenance

Despite the advancements of various methods in axle bearing fault diagnosis, several challenges remain. One significant issue is the sensitivity of fault detection techniques to noise interference in railway environments, as indicated by studies [40,62,81]. Scalability is another concern, as AI-driven models require large amounts of training data, which may not always be available in railway-specific scenarios [37, 61]. Furthermore, generalization remains a challenge, as many AI-based methods perform well on controlled datasets but struggle with unseen operating conditions and multi-fault scenarios [19,22,34].

Real-time implementation of these techniques is also an area requiring further research. While some hybrid AI models have been integrated with edge computing and IoT-enabled predictive maintenance systems [11,43,60], real-time low-latency decision-making remains a challenge. Future research should focus on enhancing fault detection reliability by improving domain adaptation techniques and integrating hybrid AI models with real-time railway monitoring infrastructure. The continued development of AI-enhanced predictive maintenance strategies, as discussed in [63,76,82], will be crucial for achieving highly reliable and efficient railway bearing diagnostics.

Fig. 6 outlines a structured methodology for developing robust and comprehensive approaches to predictive analysis and predictive maintenance. This methodology consists of three key stages, each addressing specific challenges in assessing bearing health.

In the first stage, controlled laboratory experiments are essential, as they provide a well-defined environment for testing new methods and approaches. Laboratory conditions enable precise evaluation of bearings, both in pristine condition and with artificially induced defects, allowing researchers to systematically study failure mechanisms. Additionally, laboratory test rigs should include bearings that have undergone operational wear [12,13,46], ensuring that the developed methodologies are applicable to real-world degradation patterns. The primary outcome of this stage is the development of an automated and reliable assessment methodology capable of determining bearing health with high accuracy.

In addition, the first stage could be further enhanced by incorporating modeling approaches. This allows for a completely controlled environment in which various test cases can be simulated and evaluated. A key advantage of modeling is the ability to generate clean

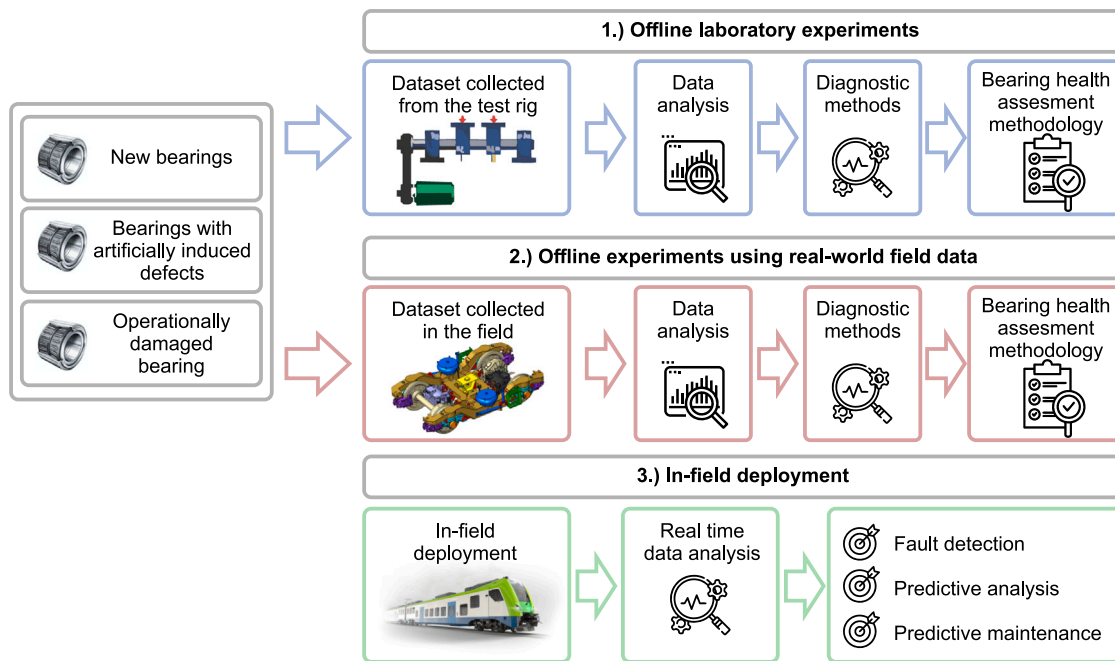


Fig. 6. Key research challenges: Developing robust methodologies for bearing health assessment, encompassing data collection from test rigs, offline analysis using in-field data, and real-time fault detection in field deployments to enhance predictive maintenance strategies.

signals that clearly indicate faults, to which noise can be systematically added [81]. This enables researchers to isolate and analyze the impact of noise on fault detection algorithms, facilitating the development of more robust and noise-resilient diagnostic methodologies.

The second stage transitions towards real-world applicability by collecting and analyzing field data. Extensive offline analysis is conducted to validate the methodology's effectiveness under conditions where data is affected by various external noise sources, such as traction motor noise and vibrations from railway operations. This stage ensures that the method remains robust when exposed to the complexities of real-world environments.

The final stage focuses on real-time in-field deployment, which presents significant technical challenges. In addition, the methodology categorizes three types of algorithms for bearing condition assessment. The first, fault detection algorithms, can evaluate the current health of the bearing but lack predictive capabilities regarding future degradation. The second, predictive analysis algorithms, extend this capability by forecasting the bearing's future condition, making them integral to predictive maintenance strategies. However, predictive maintenance itself is closely tied to railway regulations and maintenance protocols, adding another layer of complexity. Demonstrating the reliability of predictive analysis for integration into predictive maintenance frameworks remains a major research challenge.

A critical gap in existing research is the incomplete alignment with this structured methodology (see Fig. 6). Most contemporary studies focus on isolated aspects rather than covering the entire methodological spectrum necessary for an end-to-end predictive maintenance solution. To the best of our knowledge, future research should aim to establish a comprehensive framework that integrates controlled test rig experiments, real-world field deployment, and a standardized methodology for bearing health assessment. Bridging these gaps will be crucial in advancing predictive maintenance towards practical implementation in railway operations.

In addition to technical challenges, deployment of axle bearing diagnostics in operational railway systems requires compliance with stringent certification standards. In Europe, the Technical Specifications for Interoperability (TSI) mandate performance, safety, and interoperability requirements for onboard monitoring systems. In contrast,

China's TB railway standards follow different assumptions and validation criteria. Harmonizing diagnostic system performance across these regulatory regimes remains a challenge. Moreover, industrial adoption often requires compliance with the International Railway Industry Standard (IRIS) Certification scheme, which emphasizes lifecycle quality assurance, maintainability, and supplier capability. Despite the relevance of these frameworks, few studies to date address their implications in the context of AI-driven or hybrid diagnostic tools.

5.2. Specific challenges of axle bearing diagnostics

Compared to general rolling bearings used in industrial machinery, axle bearings in railway vehicles present unique technical challenges for fault diagnosis. These challenges arise from both the mechanical environment and system-level constraints:

- **Harsh operational environment:** Axle bearings are exposed to extreme dynamic loads, shocks from rail-track interactions, and variable speeds. These conditions generate complex vibration signatures with high noise levels, making early-stage fault features difficult to extract [19,81].
- **Limited sensor placement and accessibility:** In contrast to stationary bearings, axle bearings are embedded in the bogie structure, limiting access for sensor mounting and reducing the signal-to-noise ratio of measured data [8,21].
- **Non-stationary and coupled dynamic behavior:** The dynamic interaction between the wheelset, suspension, and track introduces non-linear and non-stationary components into the vibration signals, which standard time-invariant signal processing methods cannot effectively handle [19,72].
- **Real-time reliability requirements:** Diagnostic systems for axle bearings must operate in safety-critical railway environments. Therefore, they require high detection reliability with minimal latency, which is challenging under limited onboard computational resources [2,60,76].

Recent studies have proposed solutions to these challenges. For instance, adaptive signal decomposition techniques such as TVFEMD

and stochastic resonance have been used to enhance weak fault signatures under strong background noise [5,33,58]. Deep learning approaches like CNNs and LSTMs have shown promise in capturing high-dimensional features even under complex dynamic conditions [1, 7,48]. Trackside monitoring systems using beamforming and acoustic emission have been deployed to overcome inaccessibility issues [12,66, 76]. Despite these advances, robust field validation remains limited, and integration into real-time predictive maintenance frameworks is still an ongoing challenge.

Another critical challenge arises from the complex excitation forces experienced by railway axle bearings, which are fundamentally different from those acting on bearings in stationary or simpler rotating machinery. Real-world railway conditions introduce dynamic inputs such as track irregularities, wheel flat spots, and polygonal wear. These phenomena generate non-stationary and nonlinear vibrations, leading to complex modulation patterns in the bearing response. For instance, Lu et al. [81] proposed a coupling model that simulates the interaction between multiple defect types and vehicle-track dynamics, revealing how fault signatures can become obscured or distorted. Similarly, Guan et al. [19] quantified how polygonal wheel wear influences internal dynamic forces in the axle bearing, demonstrating the need for fault diagnosis techniques that can accommodate such complex, coupled excitations. Further supporting this, Wang et al. [83] investigated the dynamic forces transmitted into the axle-box bearing under polygonal wear conditions and highlighted how these excitations contribute to complex vibration patterns that challenge conventional diagnostic approaches.

5.3. Emerging trends in axle bearing diagnostics

Recent research reveals several emerging trends that signal a shift toward more integrated, intelligent, and practical diagnostic frameworks for axle bearings in railway systems.

- **Hybrid Diagnostic Frameworks:** There is a clear movement toward combining classical signal processing techniques (e.g., time-frequency decomposition, morphological filtering) with AI-based classifiers and simulation models. This hybridization allows for enhanced fault feature extraction and classification even under real-world noise and dynamic variability [30,43,58].
- **Semi-Supervised and Unsupervised Learning:** To address the lack of labeled data in real-world railway deployments, recent studies have started incorporating semi-supervised and unsupervised techniques such as One-Class SVM, autoencoders, and improved Gaussian Mixture Models [47,48]. These models improve generalization and reduce the dependency on expensive, labeled datasets.
- **Field-Oriented Signal Enhancement:** Techniques like stochastic resonance, beamforming for acoustic emissions, and noise suppression based on domain-specific knowledge (e.g., angle-domain averaging) have gained attention for mitigating interference in field environments [12,33,63].
- **Real-Time Embedded Deployment:** Lightweight models suitable for low-power hardware and edge devices are increasingly being proposed. Examples include simplified CNNs, low-resolution autoencoders, and feature selection schemes optimized for onboard diagnostics with limited computational resources [7,56,65].
- **Multimodal Sensing and Data Fusion:** The integration of temperature, vibration, acoustic emissions, and mechanical stress sensing provides a more robust picture of bearing health. Multi-task learning and fusion strategies enhance classification accuracy under variable operating conditions [12,31,32].
- **Standardization and Benchmarking:** There is increasing emphasis on standardized test protocols and publicly available datasets collected from real railway operations to facilitate benchmarking and reproducibility [18,24]. This addresses concerns about the generalizability of laboratory-developed algorithms.

- **Predictive Maintenance Integration:** Finally, efforts are underway to embed diagnostic systems into broader predictive maintenance frameworks. This includes forecasting the remaining useful life (RUL) of bearings and aligning predictions with existing railway maintenance schedules and regulations [11,27,69].
- **Simulation-Augmented Learning under Data Constraints:** Recently, simulation-based data augmentation has emerged as a promising solution to address small sample sizes and missing fault types. For instance, Bi et al. proposed a simulation data-driven fault diagnosis method for metro traction motor bearings using deep learning models trained on synthetic datasets generated under constrained real-world conditions [84].

6. Conclusions

This review has systematically analyzed 75 studies focused on predictive diagnostics of axle bearings in railway systems. It presents a unique multi-perspective classification that bridges classical signal processing, machine learning, and hybrid approaches. The review highlights the relative strengths, weaknesses, and deployment readiness of each method and identifies gaps that limit current real-world applicability.

A recurring limitation is the dependency on laboratory setups, which fail to account for real-world noise, accessibility challenges, and operational variability. Moreover, many AI models suffer from poor generalization when applied beyond the datasets they were trained on. These challenges hinder their practical adoption in predictive maintenance frameworks.

Future research should focus on the following key areas:

1. **Establishing standardized test protocols** that combine laboratory and field validation under realistic noise and load conditions to ensure diagnostic methods are applicable to real-world environments.
2. **Creating open-access, labeled datasets** collected from operational axle bearing monitoring systems, enabling the development and benchmarking of machine learning models.
3. **Developing lightweight, real-time diagnostic models** that are compatible with embedded railway platforms, addressing constraints related to onboard computational resources and latency.
4. **Exploring multi-modal sensing strategies**, including vibration, acoustic emissions (AE), temperature, and mechanical stress, to improve robustness and fault classification accuracy under variable operating conditions.
5. **Integrating diagnostics with railway maintenance decision systems**, ensuring that predictive outputs align with regulatory requirements and maintenance protocols for actionable implementation.

By addressing these challenges, axle bearing diagnostics can evolve into a more effective predictive maintenance tool, reducing unplanned downtimes, lowering maintenance costs, and significantly enhancing railway safety. This review provides a foundation for advancing real-time monitoring solutions and shaping the next generation of railway condition-based maintenance systems.

This study has several limitations that must be acknowledged. Most diagnostic methods analyzed in this review have been validated primarily in controlled laboratory environments, limiting their applicability in real-world railway operations. The transition from test rigs to operational deployment remains a challenge due to variations in environmental noise, operational conditions, and sensor placement. AI-based diagnostic methods, while promising, require large labeled datasets for effective training, and their performance in real-time railway applications remains uncertain. Additionally, the lack of standardized validation methodologies makes it difficult to compare different approaches and assess their practical effectiveness. The real-time deployment of

predictive diagnostics is also constrained by computational demands and integration challenges with existing maintenance frameworks. Addressing these limitations will be essential for improving the reliability and scalability of axle bearing diagnostics in railway applications.

CRedit authorship contribution statement

Jaromir Konecny: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Stepan Ozana:** Writing – original draft, Resources, Investigation, Funding acquisition. **Jan Choutka:** Investigation, Formal analysis. **Michal Prauzek:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT by OpenAI in order to improve the clarity and grammar of the manuscript draft. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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